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Article

Designing and Learning With Pedagogical Agents: An Umbrella Review

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Abstract

Pedagogical agents are virtual characters that instructional designers include in learning environments to help students learn. Research in the area has flourished for thirty years, yet there are still critical questions about the efficacy of pedagogical agents for influencing learning and affect. As such, we conducted an umbrella review to synthesize the field. We located 17 systematic reviews or meta-analyses focused on the use of pedagogical agents in educational settings. We found that agents can have small positive effects on learning, motivation, and other affective variables. However, we still cannot say how one should design a pedagogical agent for any given educational context. We highlight the limitations of existing theory in the area, as well as existing reviews from a practical and methodological perspective, and highlight productive areas for future research.

Keywords
pedagogical agent, virtual human, review, learning, motivation

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Introduction

For the last three decades researchers have investigated the use of virtual characters, or pedagogical agents (PAs) within multimedia learning environments to help students learn (Siegle et al., 2023). Researchers have examined the use of PAs in various contexts. For example, Abdelghani et al. (2023) designed an agent that utilized a large-language model, Jaques et al. (2023) used a PA to support collaborative learning, and Liu et al. (2022) used a PA to support meditation. To date, it is largely believed that pedagogical agents can be effective learning tools, but the results often have been mixed due to contextual differences within studies (Castro-Alonso et al., 2021; Heidig & Clarebout, 2011; Schroeder et al., 2013).

While this ever-growing body of research has provided us with vast knowledge about PAs, we still lack the ability to provide evidence-based, generalizable design principles for pedagogical agents given any particular participant population, intervention type (e.g., game-based environment, lecture video, etc.), or instructional media (e.g., virtual reality, computer-based environment, etc.). This can be problematic as PAs are continually integrated into educational technologies. Despite this limitation, the use of PAs continues to be a common research topic and there have been a number of recent

reviews attempting to synthesize the literature (Armando et al., 2022; Davis et al., 2023; Peng & Wang, 2022; Wang et al., 2023). Yet any systematic research synthesis is constrained by its inclusion and exclusion criteria,the result of which is that while we may have many reviews in the field of PAs, we have no meta-synthesis providing overarching insights into the effects and design of PAs. Accordingly, the purpose of this study is to conduct an umbrella review, also known as a review of reviews, to determine what research has been synthesized, what the findings of various reviews mean together, and extract practical implications for PA design. In addition, we document the limitations of the field that emerge through the umbrella review and highlight productive directions for future research.

Literature Review

As PA researchers, we are often asked a direct and rather blunt question: why might PAs facilitate learning? The literature provides no definitive answer. For example, some may suggest it is the pedagogy PAs can embody (Dinçer & Doganay, 2015), while others maintain it is the social processes within the learner that their presence in the learning environment initiates (Wang et al., 2008). Presumably the answer lies somewhere in between these two different conceptual positions – PAs can facilitate a variety of instruction techniques that are long established for aiding learning, such as visual signaling (Choi & Clark, 2006). However, what may provide advantages over other types of multimedia learning is their ability to initiate social processes while supporting evidence-based instructional techniques. As such, in this section we explore why social cues may be important in multimedia learning, how they can support the learning process, and how PAs can specifically embody

social cues within multimedia learning. We then explore the various aspects researchers must consider when designing PAs and summarize the diverging research findings in the area.

The Significance of Social Cues in Multimedia Learning

A frequent use of PAs in the literature is as a lecturer or an instructor to facilitate the learning process (Schroeder & Gotch, 2015). According to social agency theory, the presence of a PA exhibiting human-like social cues allows users to adopt a social stance towards the computer-generated character. The auditory and visual representations of PAs enable users to engage within a social interaction framework, leading to deeper learning (Mayer et al., 2003). This deeper cognitive engagement is due to the PA being perceived as another entity in a communicative interaction (Atkinson et al., 2005; Lee & Nass, 2003). However, researchers have shown that the qualities of the PA may determine the extent to which learning is supported by PAs. For example, human-like functionalities, such as eye gaze, facial expressions, body movement, and gestures, contribute to anthropomorphizing the PA, thereby maintaining deep user engagement (Woo, 2009). As a result, designing PAs capable of eliciting social responses can influence the cognitive and affective processes considered crucial for learning.

While social agency theory has provided a theoretical foundation for PA research for twenty years, there are few meaningful extensions of the theory beyond its initial implication that social cues can support learning. Researchers have explored how social cues can support learning. For example, Domagk (2010) found that the

valence of social cues can influence learning; however, there are few explorations based on social agency theory of how PAs influence measured social processes. This could be due to the relative lack of specific hypotheses as to which social processes influence which specific parts of the learning process. For example, from a cognitive theory of multimedia learning theory (Mayer, 2014b) perspective, how and to what extent would social processes influence the selection of relevant information, the organization of that information, or the integration of that information with prior knowledge? In an effort to address some of these theoretical questions, Schneider et al. (2022) proposed the cognitive-affective-social theory of learning in digital environments (CASTLE) that provides more specific hypotheses of how social processes may influence multimedia learning.

Cognitive-Affective-Social Theory of Learning in Digital Environments (CASTLE). The CASTLE argues that learning in digital environments inherently involves social processes that influence learning (Schneider et al., 2022). Building off a foundation of the cognitive theory of multimedia learning (Mayer, 2014b) and the cognitive-affective theory of learning with media (Moreno, 2006), CASTLE integrates social behaviors into a comprehensive theory that considers cognitive, affective, and social processes related to learning in digital environments. Based on the assumption that learners' long-term memory activates social processes when exposed to social cues, CASTLE proposes that learning can be enhanced through social facilitation in multimedia learning settings. Specifically, Schneider et al. (2022) proposed that integration of social entities into digital educational materials can boost learner motivation and foster positive shifts in attitudes towards these resources.

Consequently, PAs can be viewed as instrumental elements that support learning outcomes.

While CASTLE provides explanations for some empirical observations and the theorized effects of social cues in various learning materials, it doesn't offer explicit predictions about the magnitude or nature of how these cues influence learning outcomes (Schneider et al., 2022). Moreover, the long-standing "computers as social actors" (CASA) paradigm (Lee & Nass, 2003) suggests that interactions between humans and technological platforms, such as computers, inherently have a social dimension, extending beyond just PAs.

Summary of Relevant Theories. In summary, social agency theory and CASTLE underscore the potential to boost learner engagement and performance through the meticulous design of PAs, presumably due to the initiation of social responses within the learner. In particular, the level of anthropomorphism embedded in the design of these agents can play a crucial role in shaping how learners perceive and engage with educational content. Research to date, to some extent, supports this hypothesis (Aljaroodi et al., 2019; Garau et al., 2003). As a result, a significant body of research has focused on refining the design of PAs to amplify the social cues they offer in multimedia learning settings, with the aim of potentially enhancing learning outcomes. However, designing pedagogical agents is not a simple process. Rather, designing an effective pedagogical agent requires an in-depth understanding of the conditions it will be used in (see the Pedagogical Agent - Conditions of Use model in Heidig & Clarebout, 2011). In the next section, we explore the multitude of factors designers must consider once they have an understanding of their contextual use case.

Designing Pedagogical Agents

The design of PAs can be critically examined and compared using the Pedagogical Agents—Levels of Design (PALD) framework proposed by Heidig and Clarebout (2011). PALD organizes PA design considerations into three hierarchical layers: (1) Global design, (2) Medium design, and (3) Detailed design.

Global Design Level. At this foundational tier, designers are presented with two primary decisions: the nature of the character (whether it's human or non-human) and its mode of representation (either animated or static). While human representations are common, there are also options to use animal or object-based entities. For instance, PAs have been designed as pets for emotional support (Chen & Chen, 2014) or even as parrots that instruct in math (Atkinson, 2005).

Medium Design Level. Heidig and Clarebout (2011) suggest that the medium level is made up of two categories of decisions: technical decisions and character choices.

Technical Decisions On the technical front, decisions split into visual and auditory considerations (Heidig & Clarebout, 2011). Visually, designers must weigh factors like the degree of realism, intricacy of animation, and whether to opt for a partial or complete portrayal of the PA. The visual spectrum can span from basic illustrations to lifelike animations where the PA's movements are fluid and natural. Emotional subtleties can be integrated to enhance the agent's lifelike appearance (Schneider et al., 2022), and animations can incorporate human-like social cues such as gestures and facial

expressions (Mayer & DaPra, 2012; Mayer & Estrella, 2014). Auditory considerations encompass voice selection and modulation, with decisions such as for the PA to convey information via spoken word or text-based dialogue (Davis, 2018), or what type of voice the agent should use (Craig & Schroeder, 2017).

Character Choices The second facet of the medium level revolves around character selection. Bendel (2003) proposes a tripartite framework for this process:

1. Determine the features or capabilities the PA should possess, as these will dictate its visual and behavioral traits.
2. Define the PA's functional role, be it as an expert, teacher, mentor, peer, or facilitator (Sikström et al., 2022).
3. Decide on a physical model that aligns with the PA's role, which can range from realistic to fictional archetypes inspired by societal, artistic, or scientific contexts.

Detailed Design Level. This most intricate level involves choices that descend from decisions made in the preceding tiers. For instance, if a human character is selected at the global level, detailed choices about age, gender, physique, clothing, and other features must be determined (Heidig & Clarebout, 2011). Furthermore, if spoken dialogue is the chosen mode of communication, phonetic nuances such as intonation, emphasis, and speech pace need to be defined at this level.

The Present Study

The academic discourse surrounding the effectiveness of PAs for influencing learning and affective outcomes is inconsistent. For example, if we examine a brief (and admittedly, incomplete)

chronology, Heidig and Clarebout (2011) found largely non-significant results, while Schroeder et al. (2013) suggested a modest yet statistically significant improvement in learning outcomes due to PAs. A few years later, a review by Martha and Santoso (2019) found that only half of the studies they reviewed indicated a positive influence of PAs on learning outcomes. Similarly, Dai et al.'s (2022) analysis highlighted inconsistent empirical data on PAs in multimedia learning environments.

To summarize, reviews to date have arrived at different conclusions, ranging from mixed results (Dai et al., 2022), to small positive effects (Castro-Alonso et al., 2021; Schroeder et al., 2013), and the conclusion that we are examining too simplistic of a question that does not reflect the realities of learning environments (Heidig & Clarebout, 2011). The field has a plethora of review studies, yet all have slightly different goals, inclusion criteria, and methodological approaches. The result has been that we have seen a variety of conclusions that are difficult to reconcile.

In an effort to synthesize the vast body of review studies in the field of PAs, we conducted an umbrella review. Umbrella reviews are reviews of systematic reviews focused on understanding the overall findings in a field around a phenomenon (Aromataris et al., 2015). They enable researchers to compare and contrast the results from various syntheses to try and draw synthesized conclusions (Aromataris et al., 2015). As such, an umbrella review is the ideal methodology to help us synthesize the considerable number of systematic reviews in the field of PA research. Using this methodology, we examine the following research questions:

1. What are the publication trends in regards to systematic reviews in the field of pedagogical agents?
2. What do we know about the impact of PAs?

3. What do we know about designing PAs?
4. How well have the existing systematic reviews in the fields been conducted and reported?

Methods

An umbrella review was conducted using reporting standards from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021) and Aromataris et al.'s (2015) guidance for planning, conducting, and reporting umbrella reviews.

Literature Search

A comprehensive search of the literature was conducted to identify existing systematic reviews and meta-analyses focused on PAs. On July 15, 2023, the following nine databases were searched: Academic Search Complete, ACM Digital Library, APA PsychInfo, CINAHL, Education Research Complete, ERIC, Google Scholar, Medline with full text, and Web of Science Core Collection. This process identified 177 studies for consideration. The search string used for each database and the number of studies identified from each database are provided in Table 1. After removing 69 duplicates using Zotero (https://www.zotero.org/), 108 studies remained for consideration. An informal literature search produced an additional six studies for consideration.

Table 1. List of Databases Searched, the Search String Used for Each Database, and the Number of Studies Identified in Each Database.

Database	Studies located
Academic search complete	15
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
ACM digital Library	4
Search string (abstract): [[Abstract: "Embodied agent*"] OR [abstract: "Animated agent*"] OR [abstract: "Pedagogical agent*"] OR [abstract: "Virtual human*"]] AND [[abstract: Review] OR [abstract: Meta-ana*]]	
APA PsycInfo	19
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
CINAHL	5
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Education research complete	17
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
ERIC	11
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Google scholar	12
Search string: Allintitle: "Embodied agent" OR "animated agent" OR "pedagogical agent" OR "virtual human" AND (review OR meta-ana)	
Medline with full text	21
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Web of science core collection	73
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Total before duplicate removal	177
Duplicates removed	69
Total after duplicate removal	108
Added via informal searches	6
For screening	114

Inclusion Criteria

In order to be included in this review, existing systematic reviews and meta-analyses in the field of PAs had to meet the following criteria:

- Studies must be systematic reviews and/or meta-analyses. Reviews that did not describe a systematic search process were excluded.
- The review must be focused on non-artificially intelligent PAs. We excluded studies reviewing artificially intelligent agents programmed to interact with users or the learning environment[1], as well as reviews of avatars that were representations of self.
- The review must provide an analysis of categorizable variables or basic descriptive statistics. In other words, novel frameworks (e.g., Morrow et al., 2021) or narrative literature reviews (e.g., Siegle et al., 2023) were excluded.
- Studies must be written in either English or Korean.

Phase I Screening

The first phase of screening was conducted by one author, who reviewed all the titles and, if not clear from the title, abstracts of each study to determine if the study met the inclusion criteria. If the title and abstract were unclear if it should be included in the analysis, it was retained for full-text examination in Phase II screening. As shown in Figure 1, 41 studies were retained for full-text review.

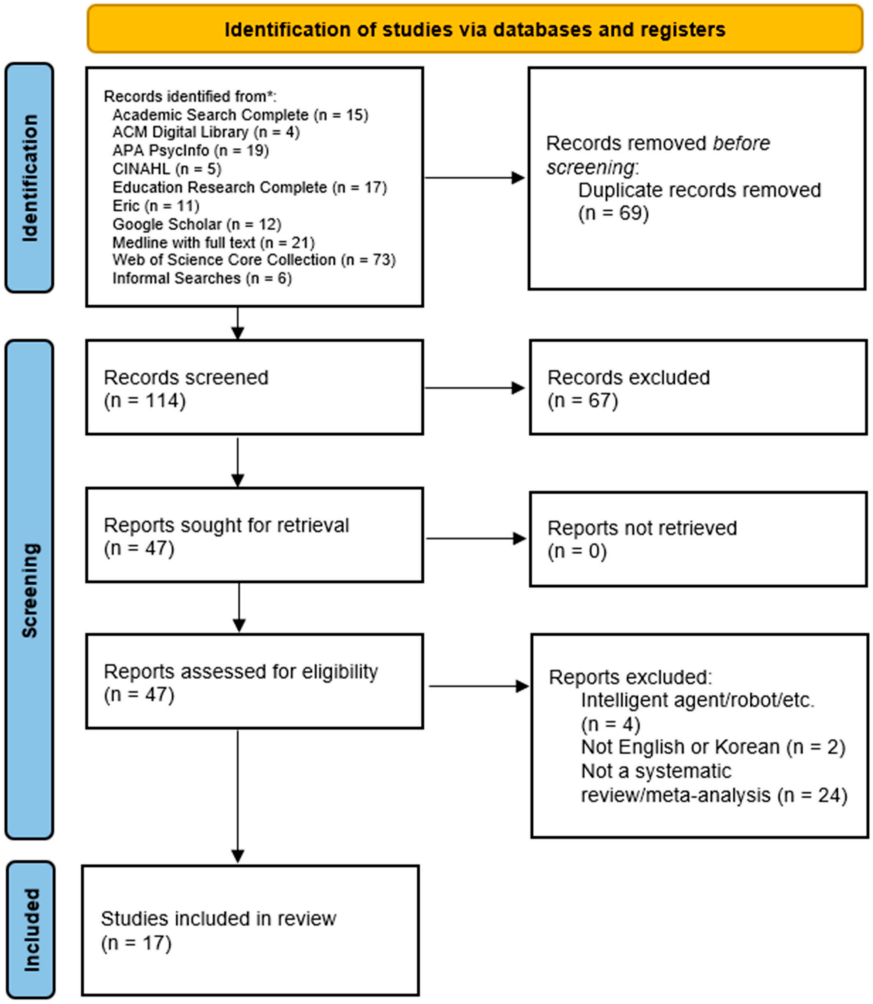

Figure 1. PRISMA flow diagram of our search and screening process (adapted from Page et al., 2021) created using Haddaway et al.'s (2022) online tool.

Phase II Screening

The second phase of screening was conducted by the same author who conducted the phase I screening, and involved obtaining the full-text copies of the retained 41 studies. The studies were read in full to see if they met the inclusion criteria. As shown in Figure 1, this process identified 17 studies that were retained for data extraction and analysis.

Data Extraction

A variety of data were extracted from each study in order to address each research question. Specifically, Table 2 specifies what information was extracted and how it was coded in respect to each research question. Our full dataset containing the codes extracted from each study is available in Supplementary Materials 1.

Table 2. The Information That was Extracted and How It was Coded in Respect to Each Research Question.

Research question	Code category	Coding scheme
1	Study	APA style citation
1	Year	Year of publication
1	Type of review	Coded as systematic review or meta-analysis
1	Analysis software	Coded as N/A (systematic review), comprehensive meta-analysis (CMA), R, or SPSS.
1	Date of database search	As reported by authors
1	Number of databases searched	As reported by authors
1	Number of articles included	As reported by authors
1	Number of participants included	As reported by authors. We coded the total number of participants across all studies if the authors reported this information. If the authors provided the number of participants in each study but did not provide a summary across all studies, it was coded as 'not reported'
1	Countries/regions of origin studies	As reported by authors
1	Quality of studies examined	Coded as the authors evaluated quality or authors did not evaluate study quality
2, 3	Research question(s) and Findings(s)	As reported by the authors. We synthesized the findings of each review in alignment to each research question. We recorded only key findings. Significant results were recorded and what was not significant was recorded when quantitative data were reported in the review. However, we did not record every single effect size reported in meta-analyses or quantitative systematic reviews
4	"Is the review question clearly and explicitly stated?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the inclusion criteria appropriate for the review question?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A

	et al., 2015, p. 136)	
4	"Was the search strategy appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the sources and resources used to search for studies adequate?" (Aromataris et al., 2015, p. 136)	Coded as yes (if more than one database was searched), no, unclear, N/A
4	"Were the criteria for appraising studies appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was critical appraisal conducted by two or more reviewers independently?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the methods used to combine studies appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was the likelihood of publication bias assessed?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were recommendations for policy/ practice supported by reported data?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the specific directives for new research appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A

Inter-Rater Agreement

Four studies (24% of the sample) were randomly selected to be coded by a second author to establish inter-rater agreement. Inter-rater agreement was found to be lower than desired (75%) when the metric for study quality was included. With study quality removed from the calculations, inter-rater agreement was substantially higher (89%). The quality evaluation metric will be addressed in the discussion section. When disagreements were found, the team referred back to the original study and discussed discrepancies until consensus was reached.

Data Analysis

Data were analyzed based on the various approaches specified below.

Research Question 1: What are the publication trends in regards to systematic reviews in the field of pedagogical agents?

This research question examines aspects such as how many reviews have been published in the area and the methods used in each study. To address this research question, basic descriptive statistics were examined.

Research Question 2: What do we know about the impact of PAs?

Researchers have examined the impacts of pedagogical agents in relation to a variety of outcome variables. As such, outcome variables from each review were categorized so that the evidence around each outcome could be synthesized.

Research Question 3: What do we know about designing PAs?

Due to the variety of details in pedagogical agent design, it was important to establish what areas reviews of pedagogical agents have been conducted. To address this research question, Heidig and Clarebout's (2011) PALD framework for pedagogical agent research was used. Study findings were organized by the levels of the PALD framework and the data were synthesized.

Research Question 4: How well have the existing systematic reviews in the fields been conducted and reported?

Examining study quality is a necessary component of the PRISMA statement (Page et al., 2021). It is also necessary to examine the quality of systematic reviews when conducting an umbrella review (Aromataris et al., 2015). However, study quality is not frequently examined in systematic reviews of educational technologies. To address this gap in the literature, we used items from Aromataris et

al.'s (2015) quality analysis instrument and examined the results descriptively.

Results

In total, 17 systematic reviews and/or meta-analyses met the inclusion criteria. Below, the results are presented in relation to each research question.

Research Question 1: What are the publication trends in regards to systematic reviews in the field of pedagogical agents?

Using the coding categories specified by Aromataris et al. (2015), the sample of systematic reviews and meta-analyses that met the inclusion criteria can be objectively described. As shown in Figure 2, while systematic reviews or meta-analyses have been consistently published in the field, the last three years have seen a dramatic increase in their number.

Figure 2. Number of systematic reviews or meta-analyses published about pedagogical agents by year. Note that our literature search was conducted in July, 2023.

Of the 17 reviews that met our inclusion criteria, seven of the 17 (41%) were meta-analyses. All of the meta-analyses were conducted using Comprehensive Meta-Analysis software (Biostat Inc, 2023). Nine of the reviews (53%) did not report when their literature searches were conducted, and authors searched between 1 and 10 databases in their literature search (mean = 5.88, median = 5). Systematic reviews and meta-analyses in the sample reviewed a minimum of 12 studies and a maximum of 115 studies (mean = 35.41, median = 28). In 10 of the 17 (59%) reviews included in our analyses the authors did not explicitly report the total number of participants in the studies they reviewed. However, those that did reported analyses with a minimum of 752 participants and a maximum of 4817 participants. Only four of the included studies discussed the countries or regions of the world in which their studies were published. However, there were varying approaches to this. For example, Armando et al. (2022) stated that they reviewed studies from "western culture", whereas *Yusoff et al. (2022) reported the number of studies by country. Finally, none of the reviews in our sample reported evaluating the quality of the studies they reviewed.

Research Question 2: What do we know about the impact of PAs?

This research question specifically focused on the use of pedagogical agents in comparison to non-agent systems (or in the case of Davis (2018) and Davis et al. (2023), the use of a static agent). Table A1 (see Appendix A) provides the summarized findings from each study in regards to learning outcomes, motivational outcomes, and other outcomes.

As shown in Table A1, ten of the reviews in our sample examined the impact of pedagogical agents on learning outcomes. Seven of the 10 studies were meta-analyses, and all showed significant, positive effects in favor of pedagogical agents. Of the three

systematic reviews, one found mostly positive results (Papoutsi & Rangoussi, 2020), while two found mostly null or mixed results (Heidig & Clarebout, 2011; Martha & Santoso, 2019).

Four of the reviews in our sample examined the impact of pedagogical agents on motivational outcomes. While the two meta-analyses found statistically significant, positive effects showing that pedagogical agents can positively impact learner motivation, the two systematic reviews did not. Specifically, Heidig and Clarebout (2011) found no convincing evidence that agents significantly impact learner motivation, and Schroeder and Adesope (2014) found a few interesting trends regarding motivation, but these were largely not supported by statistically significant results.

Finally, three reviews examined the impacts of pedagogical agents on other types of outcomes. Specifically, Wang et al. (2023) found that affective pedagogical agents can lead to significant improvements in learners' positive emotions. Meanwhile, Schroeder and Adesope (2014) and Davis (2018) examined the influence of pedagogical agents on learners' perceptions of agent persona and cognitive load. Results indicated that learners may be positively influenced by pedagogical agents in relation to their ratings of the agent's persona. Regarding cognitive load, Davis (2018) found that gestures provided minimal benefits, but did not significantly reduce cognitive load. However, Schroeder and Adesope (2014) observed a positive influence on cognitive load when the control group was a text-only learning condition, but the results were mixed when the control group was a voice-only learning condition.

Research Question 3: What do we know about designing PAs?

As shown in Table A2, 13 of the 17 reviews (76%) in our sample addressed the impacts of pedagogical agent design in their findings.

We first organized these findings according to Heidig and Clarebout's (2011) PALD framework.

The global level of Heidig and Clarebout's (2011) addresses the question of if a PA should appear as a human or some other type of character. Four of the reviews in our analysis evaluated this question. However, the results were mixed. Two of the reviews found that the agent's appearance style did not make a significant difference (Peng & Wang, 2022; Schroeder et al., 2013). However, one of the other reviews found that humanoid agents that gestured decreased cognitive load, while non-humanoid characters significantly increased cognitive load (Davis, 2018), and the other found that virtual agents were more effective than human instructors for supporting positive emotions and learners' intrinsic motivation (Wang et al., 2023).

Twelve of the 17 reviews (71%) discussed findings in relation to the medium level of pedagogical agent design, which deals with the technical decisions and specific functions or roles of pedagogical agents within the learning environment (Heidig & Clarebout, 2011). While there are a variety of different design factors that can be considered at this level, the results show a few patterns in the literature. First, there seems to be some evidence among reviews that agents should be anthropomorphized in regards to their use of nonverbal social cues (Aljaroodi et al., 2019; Dai et al., 2022). However, this is difficult to reconcile with the findings of reviews that examined the impact of gestures. For example, while three reviews (Davis, 2018; Davis et al., 2021, 2023) found that gesturing agents improved learning, two other meta-analyses did not find gesturing to be a significant moderator with respect to learning (Castro-Alonso et al., 2021; *Yusoff et al., 2022). Similarly mixed results were found in regards to facial expression and emotions. Davis et al. (2021) found that facial expressions improved learnings ratings of the pedagogical

agent's persona in most cases, yet Castro-Alonso et al. (2021) and Peng and Wang (2022) did not find that facial expressions significantly moderated agent's impacts on learning in their meta-analyses. As a whole, our analysis indicates that the data and findings from design decisions at the medium level are mixed and often contradictory. Yet, it is important to remember the context of the reviews. For example, Castro-Alonso et al. were comparing PA to non-PA systems. Their finding that facial expressions did not moderate learning outcomes is not directly comparing a PA with expressions to a PA without expressions, but rather a PA with expressions compared to a non-agent system, compared to a PA without expressions compared to a non-agent system.

Finally, four of the 17 reviews (24%) examined the detail level of the PALD framework, which focuses on the design of specific visual components of the agent such as its age or clothing (Heidig & Clarebout, 2011). From this data we also find mixed results. For example, Castro-Alonso et al. (2021) found that the agent's gender did not significantly moderate learning outcomes. However, Armando et al. (2022) argued that gender stereotypes exist when learners interact with agents. That said, their results also showed mixed results in regards to the impact of an agent's gender on learning outcomes. There are other aspects at the detail level of pedagogical agent design that were also addressed by reviews in the sample. One review addressed the issue of an agent's age. *Yusoff et al. (2022) found that the agent's age did not make a significant impact on cognitive load. In addition, one review had implications for the detail level of pedagogical agent design from a broader perspective. Aljaroodi et al. (2019) argued that it may be important to have similarities between the agent and the learner, an argument

reminiscent of the similarity-attraction hypothesis (Moreno & Flowerday, 2006).

Research Question 4: How well have the existing systematic reviews in the fields been conducted and reported?

The quality of the existing systematic reviews was examined using a metric by Aromataris et al. (2015). The metric examines 10 different aspects of systematic review and/or meta-analysis reporting.

First, it was examined if the reviews in the sample clearly and explicitly stated the research question(s) being investigated. The majority of the studies (14/17) explicitly stated the research question(s); however, three studies did not. Next, an assessment regarding the appropriateness of the inclusion criteria for answering the research questions was conducted. All reviews except one provided clear inclusion criteria to address the stated research questions.

In regards to the search strategies used; all the reviews used what could be considered an appropriate search strategy. However, two of the reviews appeared to use a limited database search, so the resources used for the search seemed inadequate.

The next examination focused on how the reviews appraised the studies they included, which can be interpreted as their data extraction method from the primary studies (i.e., the coding paradigm). Generally, these criteria were appropriate, with only three studies having unclear appraisal criteria.

However, these relatively positive evaluations do not extend further to the methods reporting of the reviews evaluated. For example, only five (29%) of the reviews had two or more independent coders extracting data. Moreover, of the seven meta-analyses, five did not explicitly address how they combined studies.

Fortunately, all but one meta-analysis did report some form of publication bias assessment.

Finally, 14 of the 17 reviews provided recommendations for policy and/or practice that were supported by their data. This left three reviews in question. Two of them were coded as "unclear" because it was not certain if the recommendations were supported by the data, and one of the reviews was coded as offering recommendations not based on its reported data. In contrast to this, all reviews suggested specific research directions, with each providing productive directions for new research.

Discussion

Research Question 1: What are the publication trends in regards to systematic reviews in the field of pedagogical agents?

The number of systematic reviews and meta-analyses in the field of pedagogical agents has increased in recent years. Regarding systematic reviews, a variety of approaches were observed, ranging from investigating impacts on learning and motivation (Heidig & Clarebout, 2011; Schroeder & Adesope, 2014) to examining specific design components, such as how pedagogical agents communicate with learners (Sikström et al., 2022). While it is important to understand how pedagogical agents broadly affect learning and motivation at the global and medium levels of Heidig and Clarebout's (2011) design framework, future research and analyses should explore design elements (detail level) of pedagogical agents and the environment. Thorough, in-depth systematic reviews in these areas may provide ample insights into how the detail level of designs impact human experiences such as perception, reception, processing, and learning.

We found that 41% of the studies in our sample were meta-analyses. Yet, all meta-analyses were conducted using the same software program, Comprehensive Meta-Analysis (Biostat Inc, 2023) versions 2 (Guo & Goh, 2015; Schroeder et al., 2013) or 3 (Castro-Alonso et al., 2021; Davis, 2018; Davis et al., 2023; Peng & Wang, 2022; Wang et al., 2023). To the best of our knowledge, this software allows for the computation of conventional meta-analyses. However, a limitation of conventional (two-level) meta-analysis is that it does not account for dependent data, so each participant's scores can only be included in one effect size. This means that groups are often excluded or combined with another group, which can dilute the specificity of the results. There are now different approaches to meta-analyses that allow for the inclusion of dependent data, namely, three-level meta-analysis. Conducting a three-level meta-analysis can be done using Simple Meta-Analysis (Schroeder, 2024) or with statistical coding such as in (R Core Team, 2022) using packages such as metafor (Viechtbauer, 2010). Fortunately, there are walk-throughs available that simplify this process (Harrer et al., 2021) and recently a three-level meta-analysis has been published in the field of educational technology that provided commented R code that could be used as a template (Schroeder et al., 2023). Therefore, meta-analysts are encouraged to explore and utilize more modern approaches to meta-analysis over conventional meta-analysis when the data and research questions support it. For example, many of the existing meta-analyses around pedagogical agents might have been better suited as three-level analyses rather than two-level (conventional) analyses.

Research Question 2: What do we know about the impact of PAs?

The results of this umbrella review showed that pedagogical agents can clearly have quantifiable benefits to learners in regards

to both learning (Castro-Alonso et al., 2021; Davis, 2018; Davis et al., 2023; Guo & Goh, 2015; Peng & Wang, 2022; Schroeder et al., 2013; Wang et al., 2023) and motivational outcomes (Guo & Goh, 2015; Wang et al., 2023). While systematic review studies found mixed results (Heidig & Clarebout, 2011; Martha & Santoso, 2019; Papoutsi & Rangoussi, 2020; Schroeder & Adesope, 2014), meta-analytic results showed statistically significant, but frequently small, benefits from the incorporation of PAs.

What is known about the impact of PAs could be the results of methodology and inclusion criteria in existing reviews. Meta-analyses quantitatively pool studies into a single effect size, potentially overlooking nuanced outcomes. Systematic reviews qualitatively assess consistencies or discrepancies but may yield inconclusive overall effects. These analyses can lead to an oversimplification of complex data. In addition, inclusion criteria can further influence results, since overlapping studies between analyses can bias outcomes. Future studies might examine the need for a mixed-methods approach with adding the qualitative analysis of systematic reviews with the quantitative assessment of a meta-analysis. Likewise, future research should investigate study overlap in meta-analyses and systematic reviews to determine if results reflect true impact or inclusion bias. We encourage readers to pragmatically evaluate outcomes, scrutinize inclusion criteria, and consider nuances in the data, especially when studies report conflicting results.

Research Question 3: What do we know about designing PAs?

By analyzing the existing systematic reviews and meta-analyses using Heidig and Clarebout's (2011) frameworks, we hoped to establish what design components are more or less effective in different contexts. However, despite the availability of numerous

reviews available investigating different layers of the framework, only two main ideas emerged from the analysis. First, in relation to the PALD framework levels of global, medium, and detail, most reviews focused on the medium level that is concerned more with technical decisions. Although some reviews measured multiple levels within a single review (Aljaroodi et al., 2019; Castro-Alonso et al., 2021; Davis, 2018; Peng & Wang, 2022; Schroeder et al., 2013; *Yusoff et al., 2022), 12 of the 17 reviews focused on the medium level. One possible explanation for the lack of review studies evaluating the global and detail levels could be the scarcity of research that evaluates those levels of design. Future experimental research should focus on the global and detailed levels of design so that future review studies can potentially provide more robust insights into those levels of design.

Secondly, a key idea explored in the medium level of pedagogical agent design was nonverbal social cues that are associated with the so-called embodiment principle. The embodiment principle states that highly embodied agents will be more effective at supporting learning than static agents (Mayer, 2014a). Since the embodiment principle does not refer to one specific aspect of PA design, but rather conflates multiple variables such as body movement, eye gaze, and gestures, we refer to these variables as non-verbal social cues. These non-verbal social cues were specifically examined in eight of the 12 studies at the medium design level (Castro-Alonso et al., 2021; Dai et al., 2022; Davis, 2018; Davis et al., 2021; Davis et al., 2023; Peng & Wang, 2022; Wang et al., 2023; *Yusoff et al., 2022). Unfortunately, there is a lack of consistent evidence in relation to specific individual cues such as eye gaze or gestures. However, eight of the 12 review studies suggest that nonverbal social cues are overall more beneficial to measured outcomes

(Aljaroodi et al., 2019; Dai et al., 2022; Davis, 2018; Davis et al., 2021, 2023; Heidig & Clarebout, 2011; Sikstrom et al., 2022; Wang et al., 2023). Only two reviews found no benefits to designing agents with nonverbal social cues (Castro-Alonso et al., 2021; Peng & Wang, 2022), and two produced mixed results (Schroeder et al., 2013; *Yusoff et al., 2022). Furthermore, when review studies are evaluated in terms of holistic nonverbal social cue strategies such as communication strategy (Sikstrom et al., 2022) and realism in behavior (Aljaroodi et al., 2019), there is a clear pattern of benefits for anthropomorphized designs that include social cues compared to disembodied conditions. In other words, there seems to be a clear benefit of including an anthropomorphized PA in the learning environment. We note however, as Heidig and Clarebout (2011) did, that there is nuance to this finding and the effectiveness of the PA will likely contextually differ and the design of the PA, including its nonverbal social cues, can influence its effectiveness (Domagk, 2010). We also note that it seems highly likely that multiple nonverbal social cues can interact with one another. For example, a facial expression may consist of mouth movements, eye brow movements, and eye movements. If any of these variables were missing or inconsistent, it may not be perceived as intended nor have the desired effect.

Research Question 4: How well have the existing systematic reviews in the fields been conducted and reported?

The findings suggested that quality of the review studies, according to Aromataris et al.'s (2015) metric, were appropriate in some respects but lacking in others. Reviews in our sample generally explicitly stated research questions, provided appropriate inclusion criteria, and used what could be considered as an appropriate database search. Moreover, all but one meta-analysis reported

publication bias, and most reviews provided policy or practice recommendations. However, using Aromataris et al.'s (2015) measure highlighted that in some reviews it was not clear if the appropriate methodologies were used. Of course, it is quite possible appropriate methodologies were used but simply not reported in the published review. For example, only five of the 17 studies documented two or more coders extracting data and only two of the seven meta-analyses explained how data was combined. Both of these points are, however, critical to understanding the trustworthiness of a review. The inter-rater agreement or reliability speaks to the reliability of the data extraction process and the understandability of the coding criteria. Meanwhile, in meta-analysis, how studies were combined can highly influence the data analysis and reported results.

Although Aromataris et al. (2015) provided a metric to evaluate the quality of review studies, we found that the evaluation can be subjective and difficult to utilize. It was challenging for our team to come to consensus on when to label something as "unclear" or "no" in some cases. For example, it is not easy to understand if inclusion criteria are appropriate or not because there are many ways to write inclusion criteria that would address the research question. Therefore, unless inclusion criteria are clearly inappropriate, a study is likely to be labeled as "yes" for this question. In addition, unless only one database was searched, what number of database searches should be conducted to be determined to be "yes" (appropriate) or "unclear?" Even though the metric lists some appropriate components that need to be considered for a review study, the subjectivity and many possibly appropriate approaches to systematic review and meta-analysis methodologies make it difficult to definitely determine what is acceptable except for when it is not addressed.

Unsurprisingly, we found that the inter-rater agreement was less than optimal when we included the ratings for each study on the quality evaluation metric. However, quality evaluation is a necessary component of umbrella reviews (Aromataris et al., 2015) and *should* be reported. In regards to systematic reviews and meta-analyses, one notable issue reviewers face is that what determines a "quality" study will likely vary by field, research question, etc. A result is that quality evaluations we have seen published in the education literature seem rather ad-hoc. This may be one reason why PRISMA moved away from using the terminology of "quality" to the terminology of "certainty" of evidence (Page et al., 2021). While we commend reviewers for attempting to create a paradigm for evaluating the quality of educational studies, it is becoming clear that the field needs some sort of guidance, similar to the PRISMA statement, for evaluating the certainty of educational studies. We therefore call for research synthesists to develop a protocol for evaluating the certainty of educational studies and reviews of educational research.

Implications for Theory

While this review was not designed to test specific theories, there are nonetheless implications for theory from this work. We found that PAs impact learning, but we still do not know how different design decisions may influence learning and why. At this point it is unclear if PAs support learning due to specific social processes during the learning process, the pedagogy they facilitate, or a combination of both factors. We postulate it is a combination of both the pedagogy and social processes, but there is limited evidence to support either perspective individually. Furthermore, it seems

plausible that personal and cultural norms may also interact with how PAs are perceived by learners, but this too is yet to be broadly investigated in the literature.

Moreover, we see ample room for research in this space around related theories and outcomes outside of learning outcomes specifically. For example, social presence is an aspect of PAs that needs to be explored beyond the mere presence or absence of an image in the design. It should focus more on how users perceive PAs in technology-mediated environments as being 'real' and how designs influence that perception (Kreijns et al., 2022). Regarding emotional benefits, earlier research explored the impact of empathetic responses and agent roles on agent design and user learning (Kim & Baylor, 2015). However, further research is needed to understand the mechanisms behind agent role design and how users recognize these roles. As a final example, consider motivational outcomes. Dai et al. (2022) provided evidence that agents can increase users' motivation, enhance agent perception, and improve learning for lower-level learning objectives. However, the lack of conclusive findings around specific design features requires a more detailed examination to understand how agent design increases motivation for educational purposes.

Taken together, our synthesis of the literature implies that social agency theory (Mayer et al., 2003) may not provide the mechanistic specificity that PA researchers require to move the field forward. However, the nonverbal social cues that make up the embodiment principle provide design elements that allow for deeper exploration in how social cues might influence user interaction with PAs. CASTLE outlines how specific social processes may influence learning, including learning from PAs. It is therefore suggested that researchers explore the use of CASTLE (Schneider et al., 2022) and

the measurement of social processes to explore the effects of both verbal and nonverbal social cues embodied by PAs on learning.

Implications for Practice

Instructional designers should take from this study a few different key points. Namely, it is clear from quantitative syntheses that PAs can positively influence learning and motivation, although the effect may not be large. What is less clear is in what situations PAs may be especially useful, or how PAs should be optimally designed for any given instructional context. As a result, we suggest that PAs be employed in situations in which they can add *educational value*. We define educational value broadly, implying positive impacts on not only learning outcomes, but also the learner's social, emotional, and motivational development. For example, PAs have been used to add educational value in ways as varied as supporting career interest (Rosenberg-Kima et al., 2008) or guiding the learner's visual attention to different areas of the screen (Johnson et al., 2013). The possibilities are endless, however much more research is needed to understand the consequences of specific design considerations. Yet, we note that PAs are not the solution to all learning problems. They are simply one educational technology tool available, and should be used when appropriate.

For systematic reviewers and meta-analysts the results of this umbrella review have clear implications as well. First, it is important that reviews are reported in accordance with the PRISMA guidelines (Page et al., 2021). We encourage fellow research synthesists to review the full PRISMA statement and the specific PRISMA extensions for different types of reviews they may be interested in. Second, as we highlighted previously, our experience highlighted that it is quite clear that educational research synthesists would

benefit from a certainty evaluation metric specifically designed for educational studies.

Limitations and Opportunities for Future Research

Our inclusion criteria required that the reviews in this analysis be of non-artificially intelligent agents. This was an intentional decision because artificial intelligence adds complexity to understanding the effects of PAs. For instance, artificially intelligent agents may provide different levels of scaffolding, switch between significantly different roles in the environment (e.g., motivator, peer, expert), or even be capable of answering or responding to students in very different ways depending on the underlying intelligence. One may be drawn to conclude that artificially intelligent agents share as much in common with PAs as they do educational chatbots. As such, it seems reasonable to view the conclusions from non-intelligent PA research, such as those presented here, as a foundation for research with intelligent PAs. For example, it seems reasonable that broad findings such as ensuring the PAs add educational value will carry over to intelligent systems, as may, for example, findings around the types of voice that the PA uses to communicate with learners. We acknowledge however, that it is possible that the results reported here may not be fully generalizable to PAs powered by artificial intelligence, so research is needed to better understand what, if any, differences there may exist.

Second, when analyzing the results of this umbrella review, we noted that relatively few reviews examined outcomes other than learning outcomes. Despite this, it is clear that PAs have the potential to significantly impact learners' motivation (Guo & Goh, 2015; Wang et al., 2023). Motivation is a complex construct and there are a variety of related variables PAs may be able to influence,

such as persistence, resilience, or self-efficacy. Primary studies indicate that agents may be able to improve these variables in different contexts. For example, Lane et al. (2013) found that a PA could improve self-efficacy for programming and Kim et al. (2007) found that a PA could influence girls' math self-efficacy. Clearly, there is promise in this area for both primary research and research synthesis.

Conclusion

We conducted an umbrella review with the aim to better understand the state of the PA literature. The goal was to explore questions such as publication trends and quality of reviews, through more specific questions such as the impact of PAs and how PAs should be designed to support specific learning processes.

Overall, the results of meta-analyses indicate that pedagogical agents seem to have consistently produced small, positive effects on learning and motivation. However, the results around designing pedagogical agents are not as clear, providing only the high-level implication that PAs should be anthropomorphized. There is still no clear guidance on how pedagogical agents should be designed to facilitate any specific outcome in any specific context. From a methodological perspective, we found that there is room for improving the reporting of existing systematic reviews and meta-analyses in the field.

In closing, while pedagogical agents seem to have beneficial effects during learning, future research needs to explore why and how. Given that large language models and generative artificial intelligence are rapidly changing the educational technology landscape, it is expected to see these technologies combined with PAs more regularly into the future. However, researchers are

encouraged to remember that while impressive technologies *can* be used for design, it is still important to understand *why* and *how* these technologies influence learning. Much more research is needed to understand why PAs have the effects they do and under what circumstances they may be most effective.

* indicates a study included in the analysis

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Note

1. We excluded artificially intelligent agents because they are often fundamentally different than the PAs investigated here in regards to their educational functions and capabilities. This is further examined in the "Limitations and Opportunities for Future Research" section.

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Appendix A

Table A1. The impact of pedagogical agents on various outcomes compared to non-agent or static agent conditions.

	Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Learning outcomes		
	Castro-Alonso et al. (2021)	Overall effect: g = 0.20, *p* = .003
	Davis (2018)	Retention: g = 0.39, *p* < .001 Near transfer: g = 0.28, *p* = .04
	Davis et al. (2023)	Multiple choice: g = 0.24, *p* < .001 Transfer: g = 0.35, *p* < .001 Cued recall: g = 0.29, *p* < .01 Free recall: g = 0.29, *p* = .04
	Guo & Goh (2015)	Retention: r = .29, *p* = .01 Transfer: r = .26, *p* < .001
	Heidig & Clarebout (2011)	No notable effects on learning
	Martha & Santoso (2019)	Only 12 (of 50) papers found significant effects from agents. 76% of papers found learning impacts, and 50% found student behavior impacts
	Papoutsi & Rangoussi (2020)	72% report positive effects on learning, 14.7% neutral, and 6% negative impacts on learning
	Peng & Wang (2022)	Overall effect of pedagogical agents: g = 0.42, *p* < .001
	Schroeder et al. (2013)	g = .19, *p* < .001
	Wang et al. (2023)	Retention: g = 0.26, *p* < .001 Transfer: g = 0.34, *p* < .001
Motivation outcomes		
	Guo & Goh (2015)	Motivation: r = .35, *p* < .001
	Heidig & Clarebout (2011)	No notable effects on motivation
	Schroeder & Adesope (2014)	"Overall, our findings indicate that pedagogical agents tend to produce more positive than negative motivational effects." p. 242 "In sum, there is not clear evidence that pedagogical agents provide any advantages for persona, motivation, or cognitive-load measures in any particular domain" p. 246 "Overall, we can see that compared to a text-only control condition, pedagogical agents may facilitate learners' perception of the systems persona, increase learner motivation, and decrease cognitive load. However, when

		decrease cognitive load. However, when compared to a voice-only condition, the results are mixed." p. 247
	Wang et al. (2023)	Intrinsic motivation: $g = 0.26$, $p < .001$
Other outcomes		
	Davis (2018)	Agent persona: $g = 0.44$, $p = .02$ Cognitive load: $g = 0.13$, $p = .16$
	Schroeder & Adesope (2014)	"Overall, it appears as though the learners' evaluation of the pedagogical agent's persona is heavily dependent on the comparison group...Thus, learners appear to prefer pedagogical agents more than non-agent multimedia learning systems, yet they may not yield the same responses as true human interaction." p.238 Results were mixed in regards to cognitive load outcomes "In sum, there is not clear evidence that pedagogical agents provide any advantages for persona, motivation, or cognitive-load measures in any particular domain" p. 246 "Overall, we can see that compared to a text-only control condition, pedagogical agents may facilitate learners' perception of the systems persona, increase learner motivation, and decrease cognitive load. However, when compared to a voice-only condition, the results are mixed." p. 247
	Wang et al. (2023)	Positive emotions: $g = 0.26$, $p = .002$

Table A2. The design implications of pedagogical agents on various outcomes.

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Global level (focusing on choosing between a human or non-human agent (Heidig & Clarebout, 2011))	
Davis (2018)	Humanoid gesturing agents decreased cognitive load while a non-humanoid character increased it
Peng & Wang (2022)	No significant moderation depending on the agent's appearance style
Schroeder et al. (2013)	No significant moderating effect depending on agent's appearance
Wang et al. (2023)	Positive emotions: Virtual agents were more effective than human instructors
	Intrinsic motivation: Virtual agents were more effective than human instructors
Medium level (focusing on technical decisions (e.g., communication techniques) and specific functions or roles of the agent (Heidig & Clarebout, 2011))	
Aljaroodi et al. (2019)	The agent should reflect human realism in behavior, appearance, and emotion. Similarity between agents and the human user may be important (not only physical appearance or gender, but also culture and ethnicity)
Castro-Alonso et al. (2021)	Appearance: 2D agents were more effective than 3D agents
	Agent gesturing, eye gaze, facial expressions, motion, and voice type were all non-significant moderators
Dai et al. (2022)	"Agents with more human-like modalities tend to perform better than less human-like ones" p. 10
Davis (2018)	Agents that use gestures can improve learning compared to non-gesturing agents: Near transfer: $g = 0.39$, $p < .001$. Gesturing agents using human voice were more effective than text-to-speech voiced agents. Retention: $g = 0.28$, $p = .04$
	Agents that use gestures can improve learners' perceptions of agent persona compared to non-gesture conditions $g = 0.44$, $p = .02$. Gesturing was more effective in shorter videos (<6 minutes) than longer videos in regards to persona ratings
	Agents that use gestures can reduce cognitive load compared to non-gesture conditions $g = 0.13$, $p = .16$. Humanoid gesturing agents decreased cognitive load while a non-humanoid character increased it
Davis et al. (2021)	Gesturing agents were rated better on facilitating learning in 5/8 studies, on credibility in 2/7 studies, human-likeness in 3/7 studies, and engagement in 4/7 studies
	Facial expressions improved the ratings on the facilitated learning subscale in 5/8 studies, and for every study for the other subscales
	Voice yielded mixed results on facilitation of learning and credibility subscales. Human voice was rated higher on human-likeness and engagement scales
Davis et al. (2023)	Multiple choice results: Full body agents were more effective than talking heads. Agents with deictic gestures had higher scores than only facial gestures
	Transfer results: Fully embodied agents were more effective than taking heads
	Free recall results: Deictic, multiple, and facial gestures were more effective than conversational or iconic gestures
Heidig & Clarebout (2011)	Explanatory feedback was better than corrective feedback to facilitate transfer performance. Agents should speak rather than provide on-screen text
Peng & Wang (2022)	No significant moderating variables (additional support, facial expressions)
Schroeder et al. (2013)	On screen text was significantly more effective than narration. Animated agents were significantly more effective than static agents
Sikstrom et al. (2022)	Pedagogical agents communicate, including interpersonally, to support self-regulation, self-efficacy, motivation, and metacognition
Wang et al. (2023)	Retention: No significant differences based on PA characteristics
	Transfer: Pedagogical agents that moved led to more learning than static agents
	Positive emotions: Multiple emotional cues were more effective than PAs with only a single emotional cue
	Intrinsic motivation: Multiple emotional cues were more effective than PAs with only a single emotional cue
*Yusoff et al., 2022	Design elements (5/17 studies): a "pedagogical agent made up with good multimedia principles give a positive result towards learner's cognitive load" p.122. No effect of cues, anxiety coping strategies or message framing
	Social cues (11/17 studies): Voice types showed mixed results. Deictic gestures, types of speech did not show significant improvement on cognitive load
Detail level (focusing on specific visual design components like the agent's age and clothing (Heidig & Clarebout, 2011))	

Aljaroodi et al. (2019)	The agent should reflect human realism in behavior, appearance, and emotion. Similarity between agents and the human user may be important (not only physical appearance or gender, but also culture and ethnicity)
Armando et al. (2022)	Gender stereotypes exist in HCI. The results were contradictory in regards to if a pedagogical agent's gender influences students' academic performance and self-evaluations The discussion suggests that agents could be used to counter/reduce stereotype threats in STEM fields The discussion suggests that a female agent may improve women's learning, interest, and self-efficacy in math The discussion suggests that the agent's gender will depend on the agent's role The discussion suggests that researchers should be careful using androgynous agents to combat gender stereotypes because the opposite of the intended effect could occur
Castro-Alonso et al. (2021)	Agent gender was a non-significant moderator
*Yusoff et al., 2022	Visual design: (1/17 studies): Age of agent had no significant impact on cognitive load

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List of Illustrations

1. Figure 1
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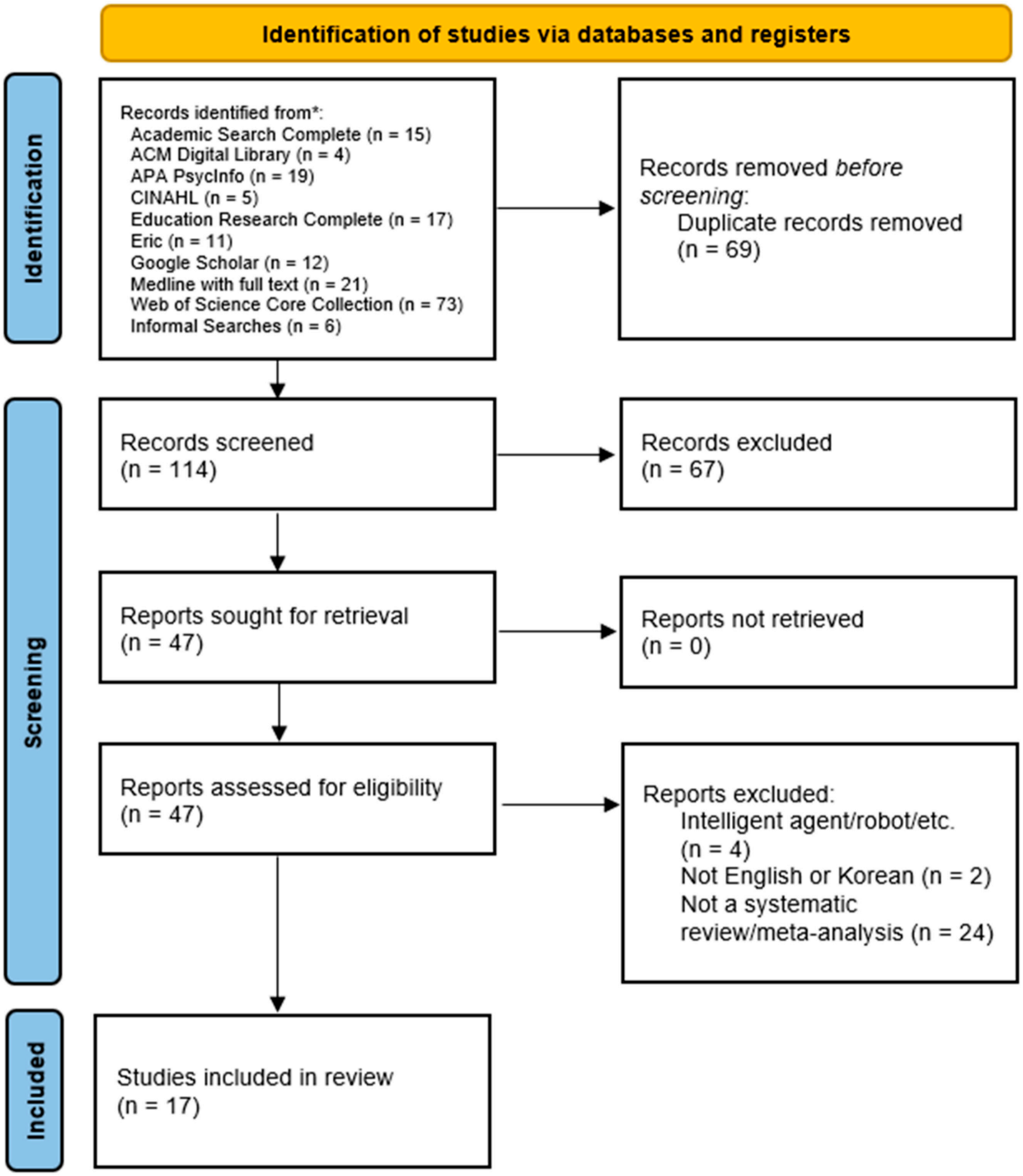

Figure 1. PRISMA flow diagram of our search and screening process (adapted from Page et al., 2021) created using Haddaway et al.'s (2022) online tool.

Figure 2. Number of systematic reviews or meta-analyses published about pedagogical agents by year. Note that our literature search was conducted in July, 2023.

Table 1. List of Databases Searched, the Search String Used for Each Database, and the Number of Studies Identified in Each Database.

Database	Studies located
Academic search complete	15
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
ACM digital Library	4
Search string (abstract): [[Abstract: "Embodied agent*"] OR [abstract: "Animated agent*"] OR [abstract: "Pedagogical agent*"] OR [abstract: "Virtual human*"]] AND [[abstract: Review] OR [abstract: Meta-ana*]]	
APA PsycInfo	19
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
CINAHL	5
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Education research complete	17
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
ERIC	11
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Google scholar	12
Search string: Allintitle: "Embodied agent" OR "animated agent" OR "pedagogical agent" OR "virtual human" AND (review OR meta-ana)	
Medline with full text	21
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	

Database	Studies located
Web of science core collection	73
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Total before duplicate removal	177
Duplicates removed	69
Total after duplicate removal	108
Added via informal searches	6
For screening	114

Table 2. The Information That was Extracted and How It was Coded in Respect to Each Research Question.

Research question	Code category	Coding scheme
1	Study	APA style citation
1	Year	Year of publication
1	Type of review	Coded as systematic review or meta-analysis
1	Analysis software	Coded as N/A (systematic review), comprehensive meta-analysis (CMA), R, or SPSS.
1	Date of database search	As reported by authors
1	Number of databases searched	As reported by authors
1	Number of articles included	As reported by authors
1	Number of participants included	As reported by authors. We coded the total number of participants across all studies if the authors reported this information. If the authors provided the number of participants in each study but did not provide a summary across all studies, it was coded as 'not reported'
1	Countries/regions of origin studies	As reported by authors
1	Quality of studies examined	Coded as the authors evaluated quality or authors did not evaluate study quality
2, 3	Research question(s) and Findings(s)	As reported by the authors. We synthesized the findings of each review in alignment to each research question. We recorded only key findings. Significant results were recorded and what was not significant was recorded when quantitative data were reported in the review. However, we did not record every single effect size reported in meta-analyses or quantitative systematic reviews

Research question	Code category	Coding scheme
4	"Is the review question clearly and explicitly stated?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the inclusion criteria appropriate for the review question?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was the search strategy appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the sources and resources used to search for studies adequate?" (Aromataris et al., 2015, p. 136)	Coded as yes (if more than one database was searched), no, unclear, N/A
4	"Were the criteria for appraising studies appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was critical appraisal conducted by two or more reviewers independently?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the methods used to combine studies appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was the likelihood of publication bias assessed?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A

Research question	Code category	Coding scheme
4	"Were recommendations for policy/practice supported by reported data?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the specific directives for new research appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A

Table A1. The impact of pedagogical agents on various outcomes compared to non-agent or static agent conditions.

	Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Learning outcomes	Castro-Alonso et al. (2021)	Overall effect: g = 0.20, *p* = .003
	Davis (2018)	Retention: g = 0.39, *p* < .001 Near transfer: g = 0.28, *p* = .04
	Davis et al. (2023)	Multiple choice: g = 0.24, *p* < .001 Transfer: g = 0.35, *p* < .001 Cued recall: g = 0.29, *p* < .01 Free recall: g = 0.29, *p* = .04
	Guo & Goh (2015)	Retention: r = .29, *p* = .01 Transfer: r = .26, *p* < .001
	Heidig & Clarebout (2011)	No notable effects on learning
	Martha & Santoso (2019)	Only 12 (of 50) papers found significant effects from agents. 76% of papers found learning impacts, and 50% found student behavior impacts
	Papoutsi & Rangoussi (2020)	72% report positive effects on learning, 14.7% neutral, and 6% negative impacts on learning
	Peng & Wang (2022)	Overall effect of pedagogical agents: g = 0.42, *p* < .001
	Schroeder et al. (2013)	g = .19, *p* < .001
	Wang et al. (2023)	Retention: g = 0.26, *p* < .001 Transfer: g = 0.34, *p* < .001
Motivation outcomes		
	Guo & Goh (2015)	Motivation: r = .35, *p* < .001

	Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
	Heidig & Clarebout (2011)	No notable effects on motivation
	Schroeder & Adesope (2014)	"Overall, our findings indicate that pedagogical agents tend to produce more positive than negative motivational effects." p. 242
		"In sum, there is not clear evidence that pedagogical agents provide any advantages for persona, motivation, or cognitive-load measures in any particular domain" p. 246
		"Overall, we can see that compared to a text-only control condition, pedagogical agents may facilitate learners' perception of the systems persona, increase learner motivation, and decrease cognitive load. However, when compared to a voice-only condition, the results are mixed." p. 247
	Wang et al. (2023)	Intrinsic motivation: g = 0.26, $p < .001$
Other outcomes		
	Davis (2018)	Agent persona: g = 0.44, $p = .02$
		Cognitive load: g = 0.13, $p = .16$
	Schroeder & Adesope (2014)	"Overall, it appears as though the learners' evaluation of the pedagogical agent's persona is heavily dependent on the comparison group...Thus, learners appear to prefer pedagogical agents more than non-agent multimedia learning systems, yet they may not yield the same responses as true human interaction." p.238
		Results were mixed in regards to cognitive load outcomes
		"In sum, there is not clear evidence that pedagogical agents provide any advantages for persona, motivation, or cognitive-load measures in any particular domain" p. 246
		"Overall, we can see that compared to a text-only control condition, pedagogical agents may facilitate learners' perception of the systems persona, increase learner motivation, and decrease cognitive load. However, when

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
	compared to a voice-only condition, the results are mixed." p. 247
Wang et al. (2023)	Positive emotions: g = 0.26, *p* = .002

Table A2. The design implications of pedagogical agents on various outcomes.

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Global level (focusing on choosing between a human or non-human agent (Heidig & Clarebout, 2011))	
Davis (2018)	Humanoid gesturing agents decreased cognitive load while a non-humanoid character increased it
Peng & Wang (2022)	No significant moderation depending on the agent's appearance style
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Wang et al. (2023)	Positive emotions: Virtual agents were more effective than human instructors
	Intrinsic motivation: Virtual agents were more effective than human instructors
Medium level (focusing on technical decisions (e.g., communication techniques) and specific functions or roles of the agent (Heidig & Clarebout, 2011))	
Aljaroodi et al. (2019)	The agent should reflect human realism in behavior, appearance, and emotion. Similarity between agents and the human user may be important (not only physical appearance or gender, but also culture and ethnicity)
Castro-Alonso et al. (2021)	Appearance: 2D agents were more effective than 3D agents
	Agent gesturing, eye gaze, facial expressions, motion, and voice type were all non-significant moderators
Dai et al. (2022)	"Agents with more human-like modalities tend to perform better than less human-like ones" p. 10
Davis (2018)	Agents that use gestures can improve learning compared to non-gesturing agents: Near transfer: g = 0.39, $p < .001$. Gesturing agents using human voice were more effective than text-to-speech voiced agents. Retention: g = 0.28, $p = .04$
	Agents that use gestures can improve learners' perceptions of agent persona compared to non-gesture conditions g = 0.44, $p = .02$. Gesturing was more effective in shorter videos (<6 minutes) than longer videos in regards to persona ratings
	Agents that use gestures can reduce cognitive load compared to non-gesture conditions g = 0.13, $p = .16$. Humanoid gesturing

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
	agents decreased cognitive load while a non-humanoid character increased it
Davis et al. (2021)	Gesturing agents were rated better on facilitating learning in 5/8 studies, on credibility in 2/7 studies, human-likeness in 3/7 studies, and engagement in 4/7 studies
	Facial expressions improved the ratings on the facilitated learning subscale in 5/8 studies, and for every study for the other subscales
	Voice yielded mixed results on facilitation of learning and credibility subscales. Human voice was rated higher on human-likeness and engagement scales
Davis et al. (2023)	Multiple choice results: Full body agents were more effective than talking heads. Agents with deictic gestures had higher scores than only facial gestures
	Transfer results: Fully embodied agents were more effective than taking heads
	Free recall results: Deictic, multiple, and facial gestures were more effective than conversational or iconic gestures
Heidig & Clarebout (2011)	Explanatory feedback was better than corrective feedback to facilitate transfer performance. Agents should speak rather than provide on-screen text
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Sikstrom et al. (2022)	Pedagogical agents communicate, including interpersonally, to support self-regulation, self-efficacy, motivation, and metacognition
Wang et al. (2023)	Retention: No significant differences based on PA characteristics
	Transfer: Pedagogical agents that moved led to more learning than static agents
	Positive emotions: Multiple emotional cues were more effective than PAs with only a single emotional cue
	Intrinsic motivation: Multiple emotional cues were more effective than PAs with only a single emotional cue

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
*Yusoff et al., 2022	Design elements (5/17 studies): a "pedagogical agent made up with good multimedia principles give a positive result towards learner's cognitive load" p.122. No effect of cues, anxiety coping strategies or message framing
	Social cues (11/17 studies): Voice types showed mixed results. Deictic gestures, types of speech did not show significant improvement on cognitive load
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Aljaroodi et al. (2019)	The agent should reflect human realism in behavior, appearance, and emotion. Similarity between agents and the human user may be important (not only physical appearance or gender, but also culture and ethnicity)
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	The results were contradictory in regards to if a pedagogical agent's gender influences students' academic performance and self-evaluations
	The discussion suggests that agents could be used to counter/reduce stereotype threats in STEM fields
	The discussion suggests that a female agent may improve women's learning, interest, and self-efficacy in math
	The discussion suggests that the agent's gender will depend on the agent's role
	The discussion suggests that researchers should be careful using androgynous agents to combat gender stereotypes because the opposite of the intended effect could occur
Castro-Alonso et al. (2021)	Agent gender was a non-significant moderator
*Yusoff et al., 2022	Visual design: (1/17 studies): Age of agent had no significant impact on cognitive load